



CropSense

*Imaging Platform for Plant
Analysis*

Phase Two

Meet the Team



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Problem Statement

The Need for Early Detection:

Multispectral, hyperspectral, and thermal imaging are highly effective for detecting subtle physiological changes in plants (e.g., *Zymoseptoria tritici* in wheat) before visible symptoms appear.

Operational Bottlenecks:

Previously, processing high-volume phenotypic data required tedious manual manipulations, leading to frequent errors and data corruption.

Project Objective:

Phase 2 of CropSense simplifies this by developing an automated, unified platform with an intuitive user interface to efficiently acquire, process, and visualize complex data.

Bridging Research and Operations:

The platform transitions raw research data into structured datasets, supporting the future development of robust predictive models for plant disease.

Methodology

Strict Multi-Sensor Acquisition Protocol:

Captures data using Hyperspectral (Specim FX10), Multispectral (Silios Toucan), and Thermal (Workswell WIRIS) cameras. Acquisition prioritizes hyperspectral/multispectral first, followed by thermal, to minimize the impact of lighting variations and plant movement.

Temporal Fusion (10-Minute Window):

Uses a synchronization algorithm to group independent camera outputs captured within ± 10 minutes of each other into a single, unified "Capture Event" for a specific plant.

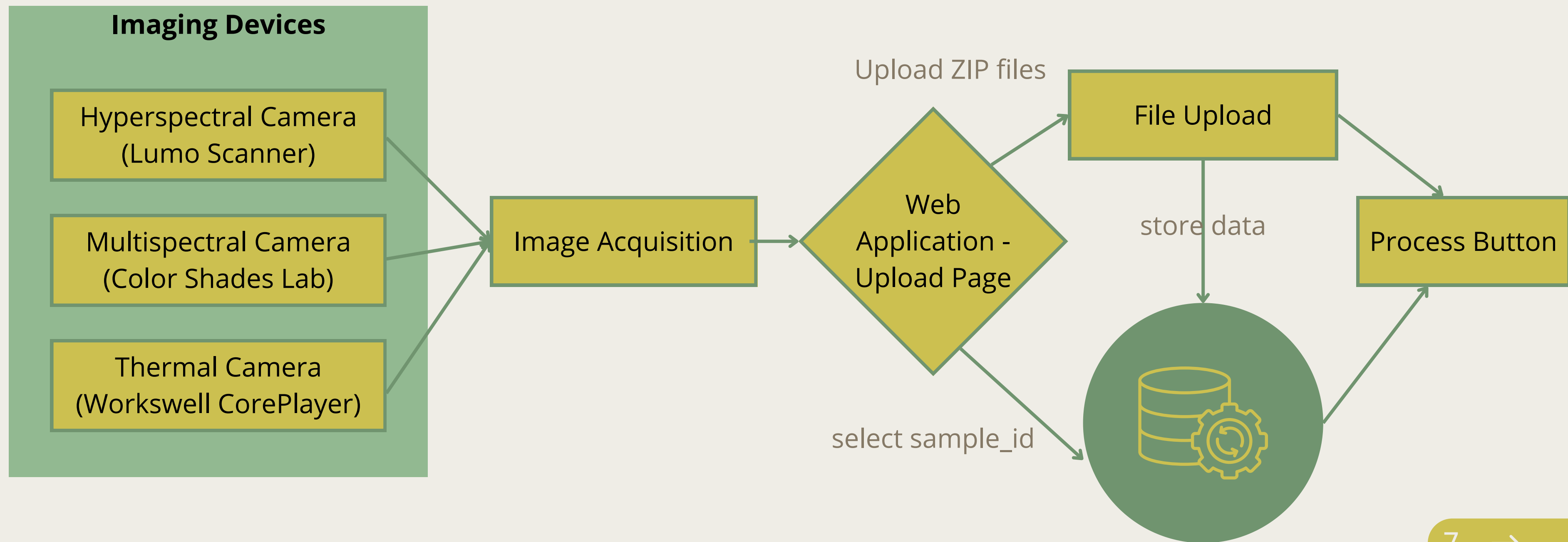
Automated 4-Step ETL Ingestion:

Compliment the manual "Drop Zone" with an automated Extract-Transform-Load pipeline featuring four steps: File Detection, Deduplication, chunked Binary Upload to MongoDB (GridFS), and Metadata Insertion.

Architecture & ML Preparation:

A Flask backend orchestrates the pipeline, automatically generating placeholder documents for new multispectral/thermal captures to signal the ML module for plant health classification.

System Architecture 1



Database Organization

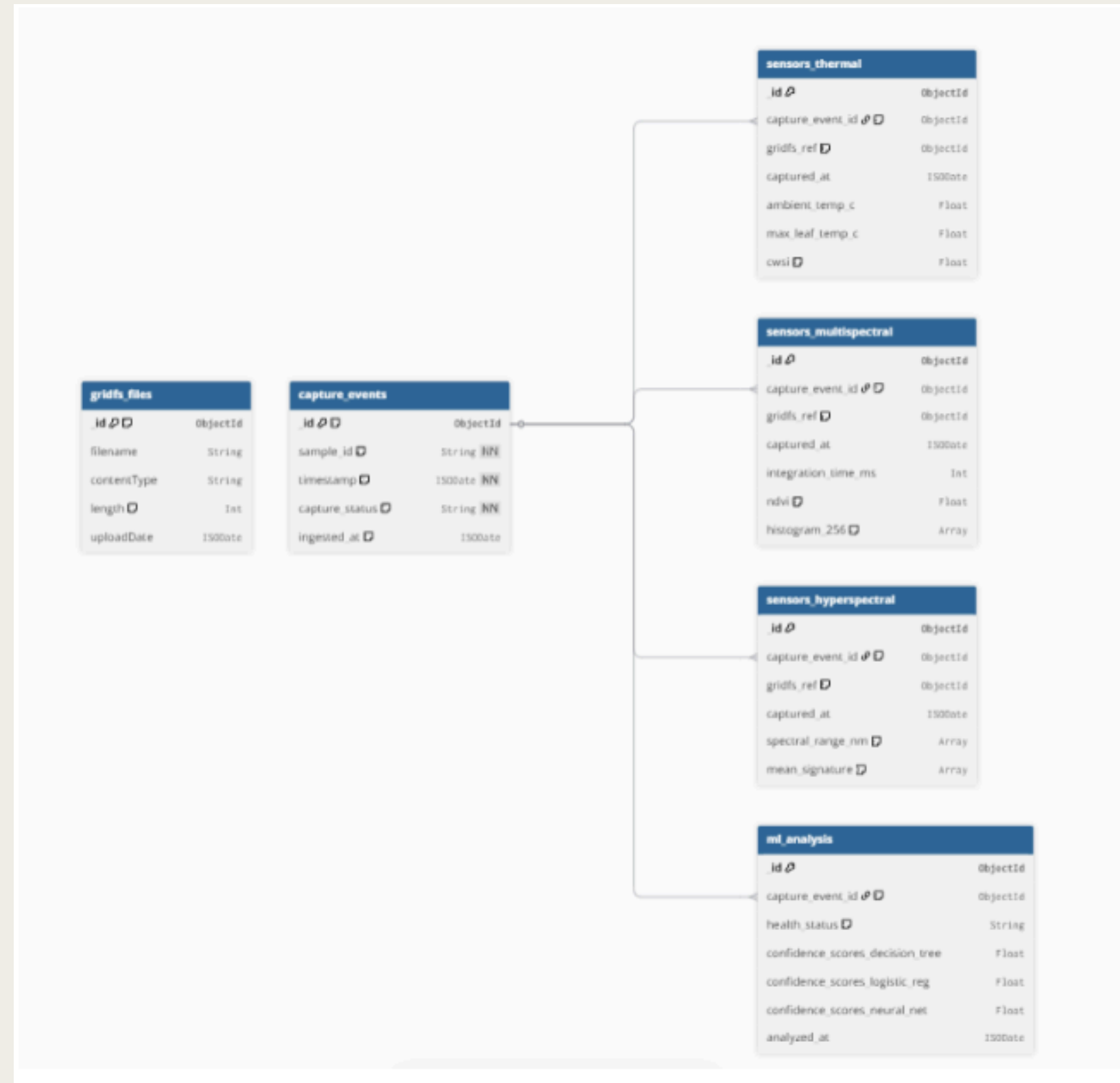


MongoDB, a NoSQL database, was chosen over relational alternatives to handle both structured scientific metadata and large binary files.

A central `capture_events` collection anchors every acquisition session using a plant `sample_id` and timestamp.

Sensor-specific metadata and machine learning results are housed in dedicated collections that reference the central event via an `ObjectId`.

The system utilizes MongoDB's native GridFS subsystem to automatically split large raw image files into 255 KB chunks without needing a separate file server.



System Architecture 2

Process Button

Select Parameters

Processing Pipeline
(ETL)

Config Page

Submit

Plant Statistics &
Visualization

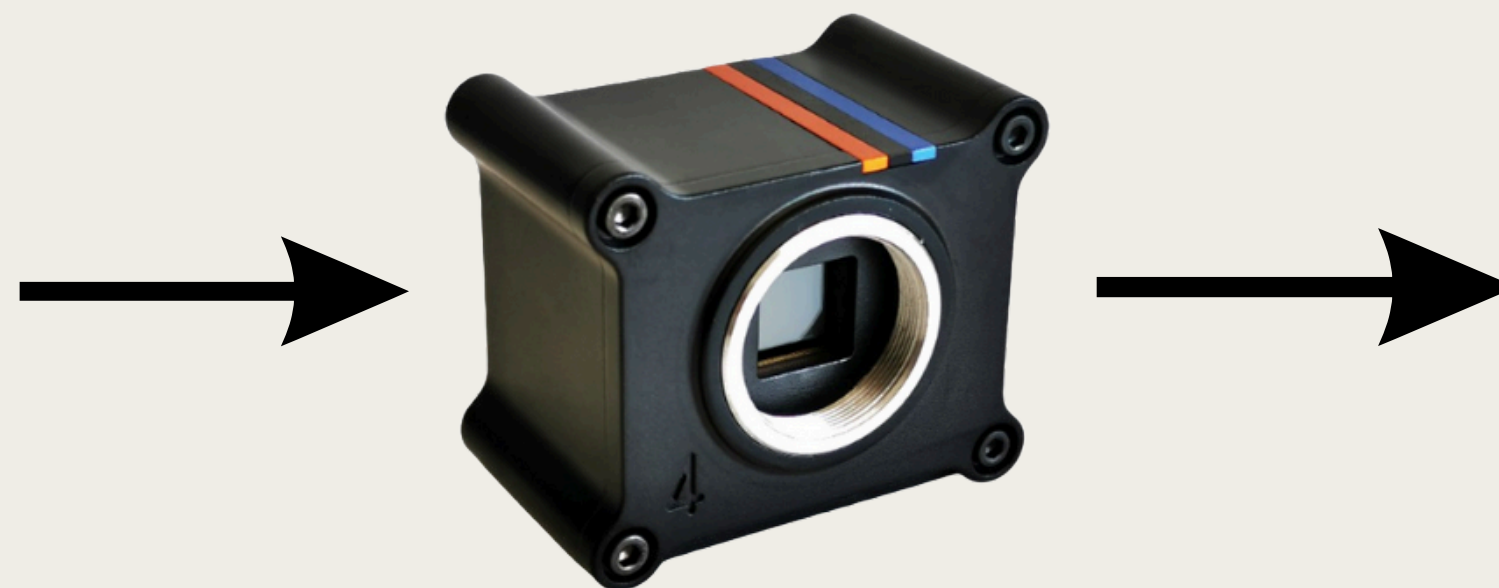


Hardware System



Hyperspectral (Specim FX10)

- Captures full spectral signature
- Line-scan (pushbroom)
- High precision, most complex setup
- Needs white & dark calibration



Multispectral (Silos Toucan)

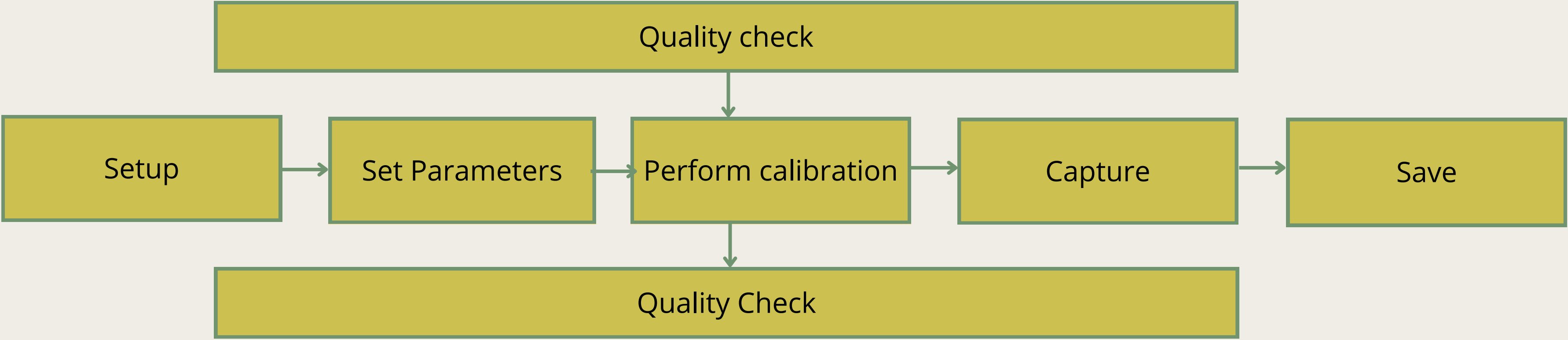
- Captures selected spectral bands
- Snapshot imaging (no scanning)
- Medium complexity
- Requires white & dark calibration



Thermal (WIRIS Agro)

- Measures temperature (infrared)
- Very simple acquisition
- Measures stress

Camera Protocol



Hyperspectral (Specim FX10)

- Scan speed: 9.0
- Positioning speed: 50.0
- Frame rate: 50 fps
- Exposure time: 8 ms
- Working distance: ~27–29 cm
- Acquisition mode: Manual

Multispectral (Silos Toucan)

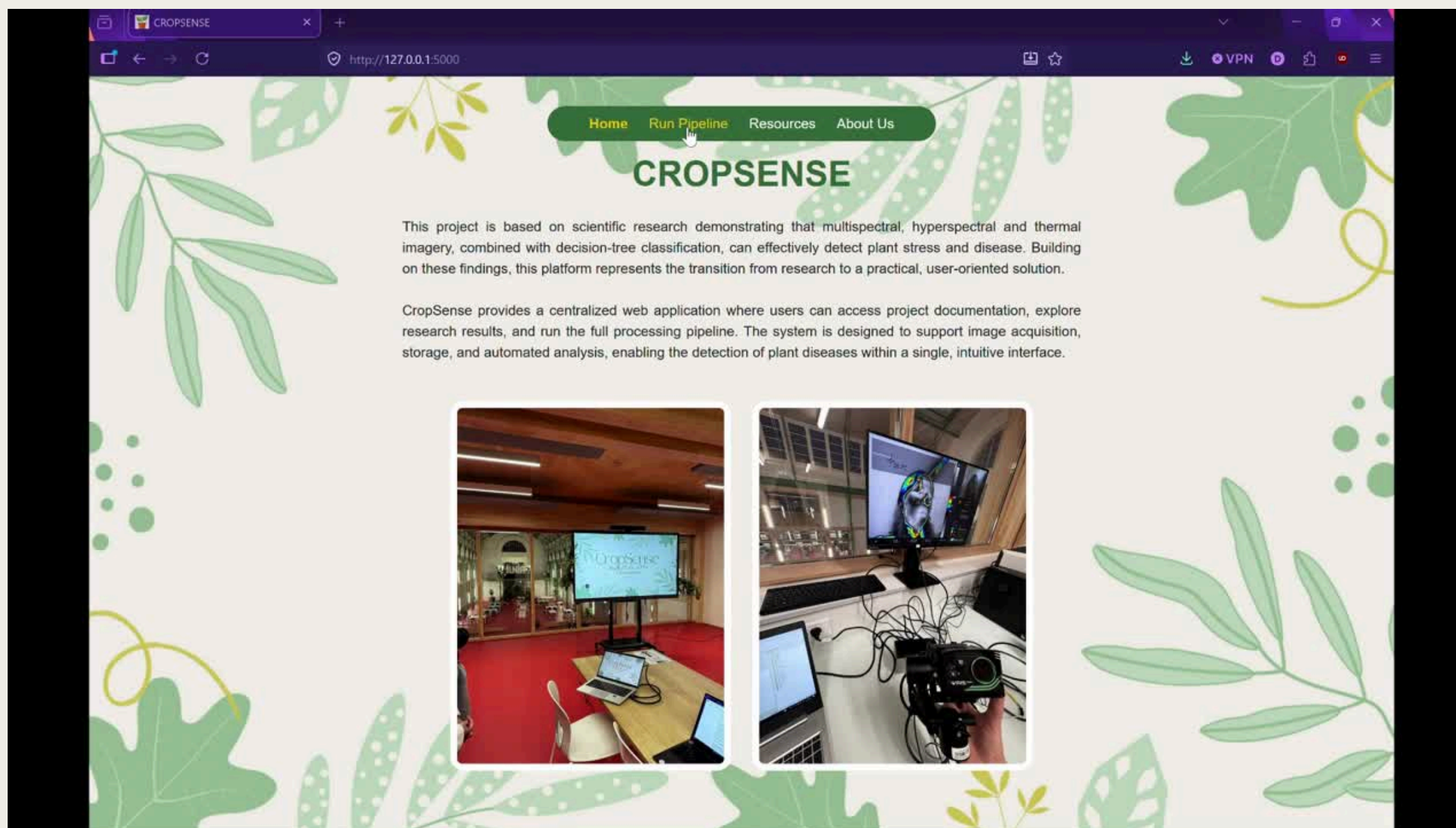
- Focus distance: ~1 m
- Aperture: f/2.8 – f/4
- Integration time: ~8.33 ms
- Frame rate: ~60 fps (software)
- Software: ColorShades Lab

Thermal (WIRIS Agro)

- Capture time: ~1 s per image
- Frame rate: ~50 Hz (video output)
- Calibration: Internal (NUC)

DEMO

<http://127.0.0.1:5000>

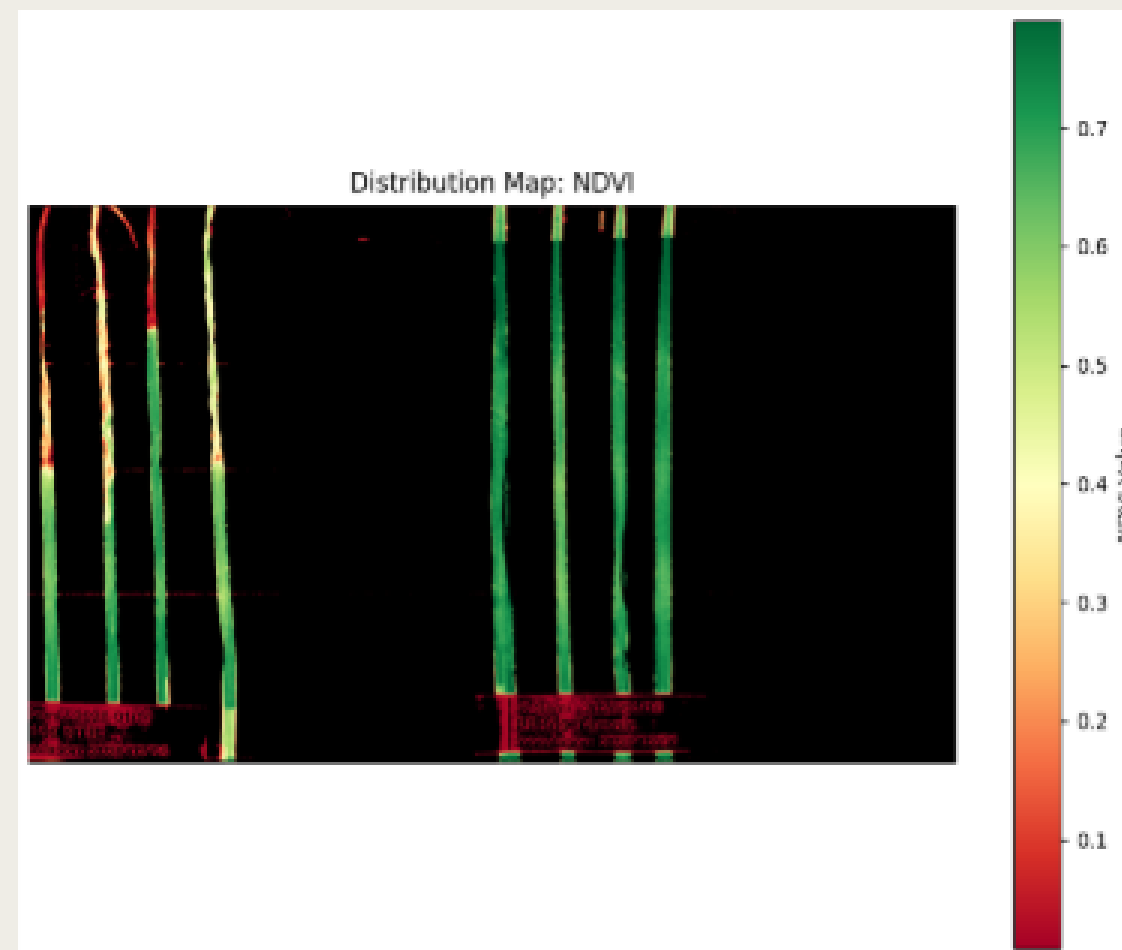
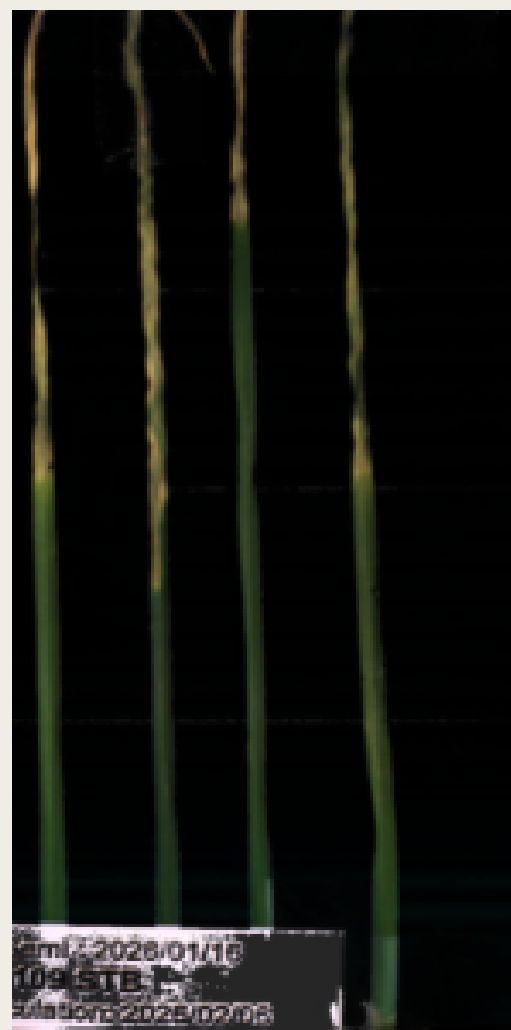


Various Visualization

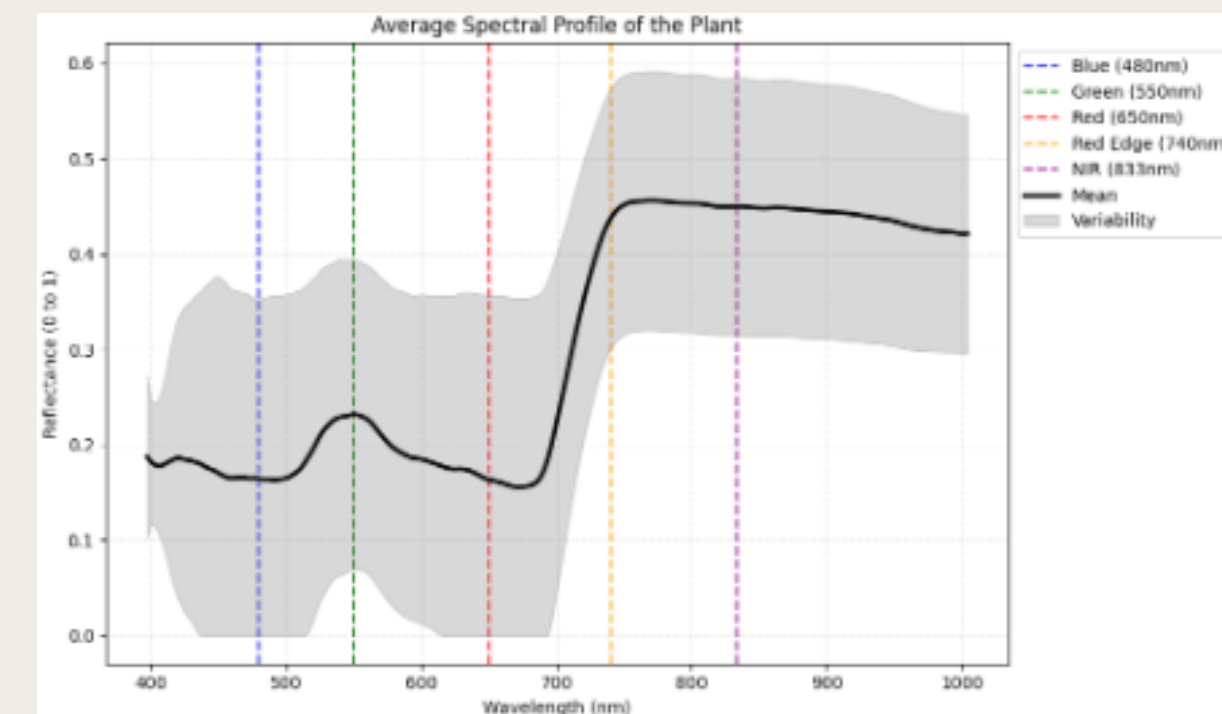
False-Color RGB Imagery:



**Spatial Heatmaps
of Vegetation Indices:**

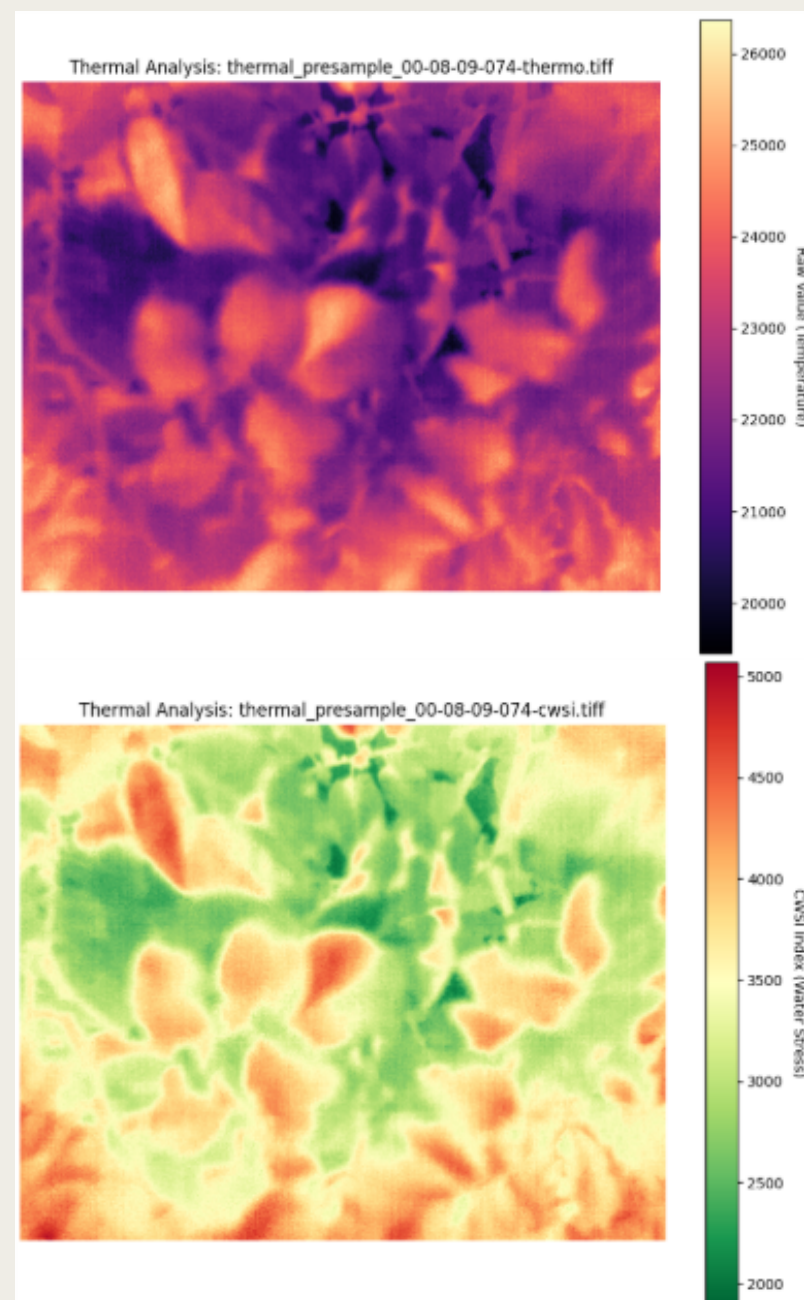


Spectral Signature Graphs:

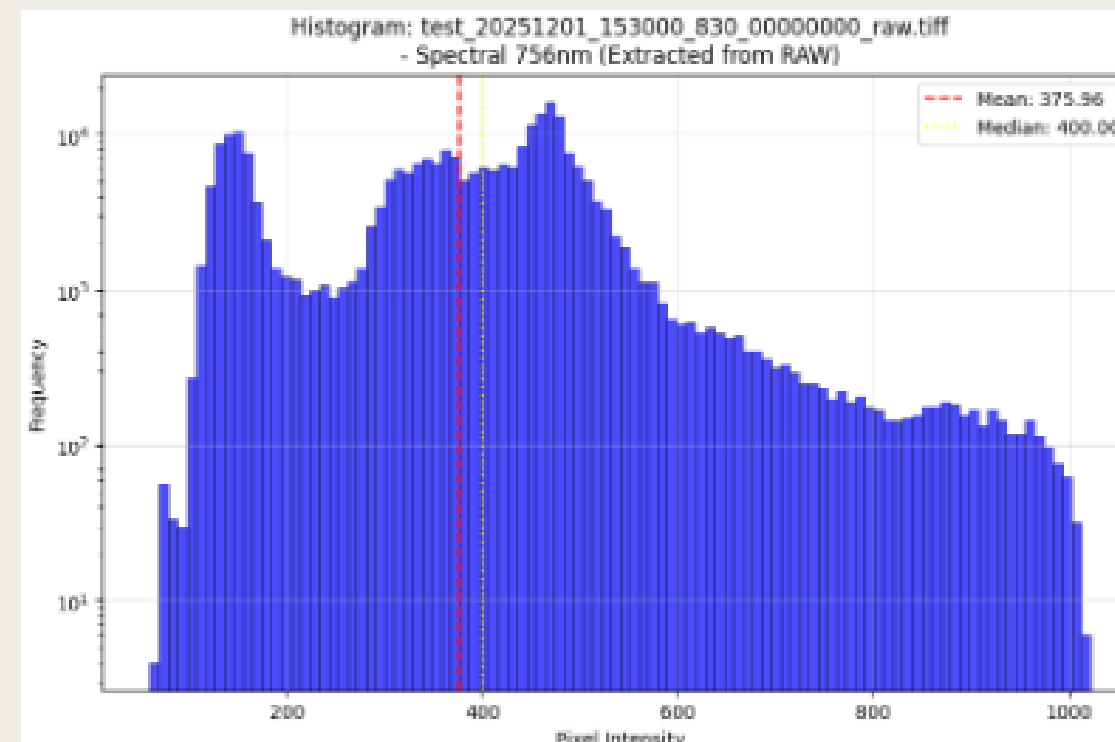


Various Visualization

Thermal and CWSI Heatmaps:



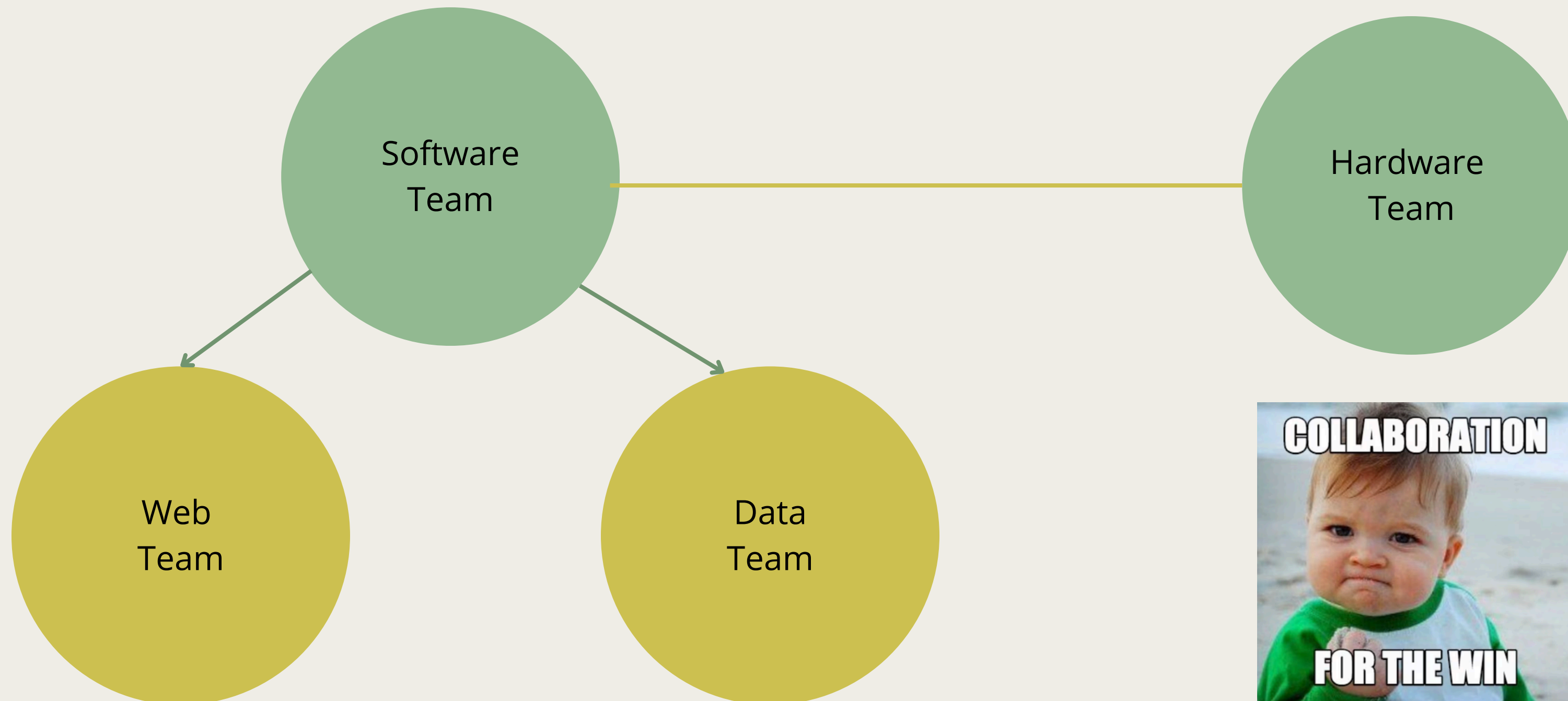
Multispectral Intensity Histograms:



Comparative Analytics Charts:

	Control Plant	Inoculated Plant
Health Indices		
NDVI (800/670)	0.58231	0.42452
GNDVI	0.36079	0.24967
RVI	2.90605	2.12684
WI	1.04077	1.03530
NDWI	0.01967	0.01649
SIPI	0.94147	1.07841
PRI	-0.02723	0.00402
ARI	1.08891	1.21597
CARI	3.53524	2.08058

Project Management



Project Management: How?



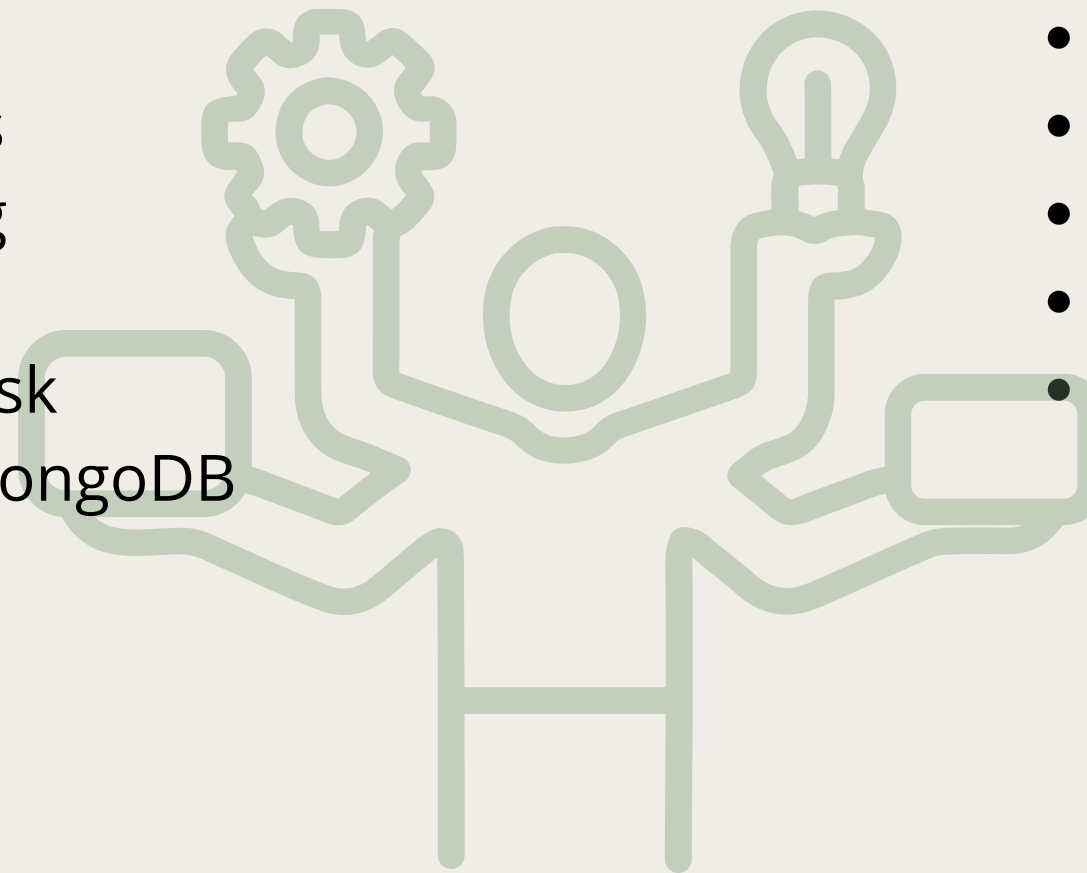
Skills ACQUIRED

Technical Skills

- Agricultural data analysis / smart agriculture concepts
- Hyperspectral imaging analysis
- Multispectral image processing
- Thermal imaging and analysis
- Backend development with Flask
- Database management with MongoDB

Soft Skills

- Communication
- Teamwork and collaboration
- Planning and organization
- Conflict resolution
- Public speaking and presentation skills
- Time management



Future Extensions - Model

What's done:

- 9 vegetation indices extracted as features (NDVI, GNDVI, RVI...)
- 5 classifiers trained on synthetic data (n=240)
- **Best model: NN medium MLP – F1 = 0.898, AUC = 0.898**
- Model serialised → ready for web app inference

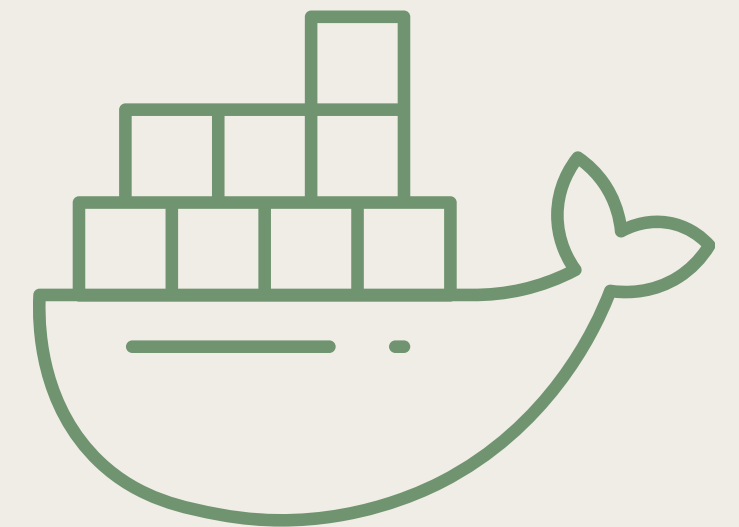
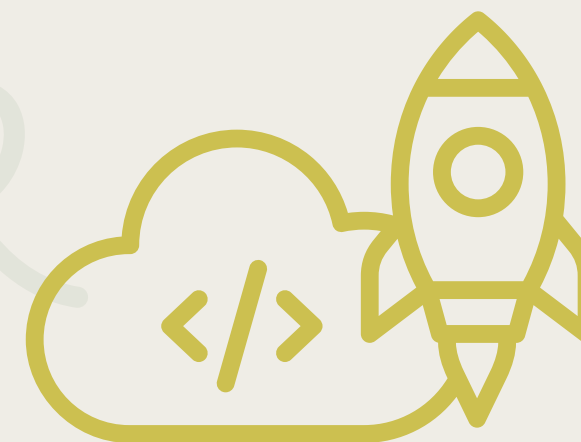
What's next:

- Collect real labelled field data (plant_features.csv)
- Add thermal features
- Integrate live prediction into web application
- Optional: add SHAP explainability for agronomists



Future Extensions - Web App

UI for model prediction



Deployment



Conclusion



Thank You